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Sizewell C
The power of **good for Britain**

SIZEWELL C (SZC) LTD

CFSI HANDBOOK

Sizewell C
The power of good for Britain

TABLE OF CONTENTS

1 FOREWORD

This guidebook is intended to promote a greater understanding of **Counterfeit, Fraudulent and Suspect Items** (thereafter **CFSI**), their potential impact and what the Sizewell C (SZC) Project expects of its supply chain partners on this important issue. This guidebook includes some real-life examples of CFSI to illustrate how CFSI has impacted supply chains.

Counterfeiting and fraud are an increasing and widespread global concern and poses serious threats to the economies and general market stability of the world. Studies show that the fundamental need for a controlled supply chain has taken on a new level of criticality.

CFSI presents a significant and specific risk to the nuclear industry and new build projects such as SZC, as recognised by the UK Nuclear Regulator (Office for Nuclear Regulation - ONR), with potential to impact upon the successful delivery of our nuclear programmes and our reputational and commercial success. CFSI also poses immediate and potential threats to worker safety, facility performance, the public, and the environment.

Nuclear facilities and their supply chains should be acutely aware of the CFSI topic and must be aware of the risks to the industry, but particularly aware to the CFSI risks posed to their specific business.

Preventing CFSI from entering the SZC Project is of paramount importance to ensure successful delivery of our new nuclear programme. Organisations across the nuclear industry supply chains must implement effective measures to prevent the introduction and use of CFSI, including raw materials, components, and documentation.

The supply chain has a key role in preventing and detecting the entry of such counterfeit or fraudulent items, or indeed any non-conforming substandard item, into nuclear facilities. This handbook is aimed at supporting the SZC supply chain understand the topic of CFSI and the risk it presents, to strengthen suppliers' arrangements in the assessment and mitigation of CFSI risks to their business, and the products and/or services supplied to the SZC Project.

The SZC aim is to support the delivery of excellence in all aspects of the Project, with a healthy Nuclear Safety Culture[1] at the forefront of our collective response to the threat of CFSI, we are all committed to achieving excellence in all our activities to ensure the safe, quality delivery of the Project and future safe plant operation.

[1] https://www.iaea.org/topics/safety-and-security-culture

2 INTRODUCTION

Many factors contribute to the growing number of CFSI entering the marketplace. CFSI may have been inadvertently procured and already installed in nuclear facilities.

The infiltration into the global supply chain of CFSI is a growing concern worldwide. Suppliers risk losing profit, intellectual property and reputation, and the risk of job losses is increased.

Experience has shown that a lack of control of the processes involved in the management of specifying, sourcing, receipt, inspection and verification, use of or disposal of items can all lead to the introduction of counterfeit or fraudulent items into a nuclear facility. This has led in some cases of plants being shut down until the impacts of such materials can be assessed and mitigated as required.

Concern about CFSI extends beyond the raw materials, components, and equipment levels. Even when equipment is purchased from the Original Equipment Manufacturer (OEM), there is a possibility that the materials or components used in its manufacture may be counterfeit or fraudulent. CFSI can also occur in an individual’s claims of experience, training, qualifications and certifications impacting their ability, authority or competence necessary to perform their role as Suitably Qualified and Experienced Personnel (SQEP).

CFSI need to be detected, identified and isolated as early as practicable, and their individual impact on safety, costs and work schedules need to be evaluated, to determine an appropriate course of action. This includes communicating and documenting information internally and sharing the resulting lessons learned with the nuclear industry.

Engineering and procurement personnel also face challenges related to the increasing number of CFSI being introduced into the supply chain. These challenges can stem from items from the original manufacturer no longer being available or the manufacturer no longer being willing to support the rigorous testing and documentation, or material certification processes needed for some items. Engineering and procurement personnel thus often rely on commercial items because nuclear grade compliant items are not available. Unfortunately, these conditions are well suited to vendors that are willing to intentionally supply CFSI to increase their profit margins.

Some vendors can take advantage of weaknesses in the processes used to prevent or address CFSI issues. There may be poorly defined procurement specifications, weak or absent procurement clauses prohibiting CFSI, insufficient depth or control of vendor qualification processes, weak or absent receipt inspection acceptance criteria, a lack of consequences if CFSI are discovered, or a lack of information sharing about non-conforming vendors within the nuclear facility supply chain, all of which can have major impacts upon operational safety and security.

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SZC places Safety & Quality at the forefront of the **Project Values** (**Fig. 1**). A robust **safety culture** throughout the organisation promotes openness and transparency for communicating potential problems and lessons learned, including the incorporation of new processes and applications.

When abnormal conditions are identified that involve a product or service, the item should be considered **Suspect** until a subsequent investigation or test demonstrates the item to be genuine, non-conforming, counterfeit or fraudulent.

The potential for CFSI to enter nuclear facilities can be reduced significantly by adopting a strong safety culture.

Figure 1. SZC Project Values

3 CFSI DEFINITION

Counterfeit

Items intentionally manufactured, refurbished or altered to imitate an original or legitimate product without authorisation, to pass themselves off as genuine. Counterfeit product can be deficient materially or have an inability to reliably function to the specified conditions.

Fraudulent

Fraudulent items are intentionally misrepresented with intent to deceive, including items with incorrect identification or false/inaccurate certifications. They may also include items sold by entities that have acquired the legal right to manufacture a specified quantity of an item but produce a larger quantity than authorised and sell the excess as legitimate inventory.

Suspect

An item about which there is an indication by visual inspection, testing, or other preliminary information that it may not conform to accepted standards, specifications and/or technical requirements and there is a suspicion that the item may be counterfeit, fraudulent or non-conforming.

Additional information or investigation is then needed to determine whether the suspect item is acceptable, non-conforming, counterfeit or fraudulent.

Item

Although **I** stands for item, it can be a product but can also be a service, documentation or certification.

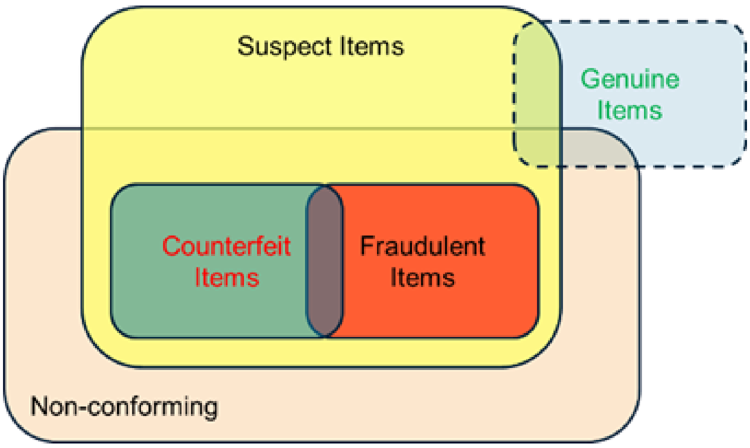


Figure 2. CFSI Definition

4 BACKGROUND

There are many factors that contribute to the ever-growing numbers of CFSI entering the marketplace.

- Counterfeiting made up approximately 2.5% of total world trade in 2021[2].
- There is no sign that the impact is going to decrease soon. In fact, the indications are that the problem is going to increase.

Counterfeiting in Perspective:

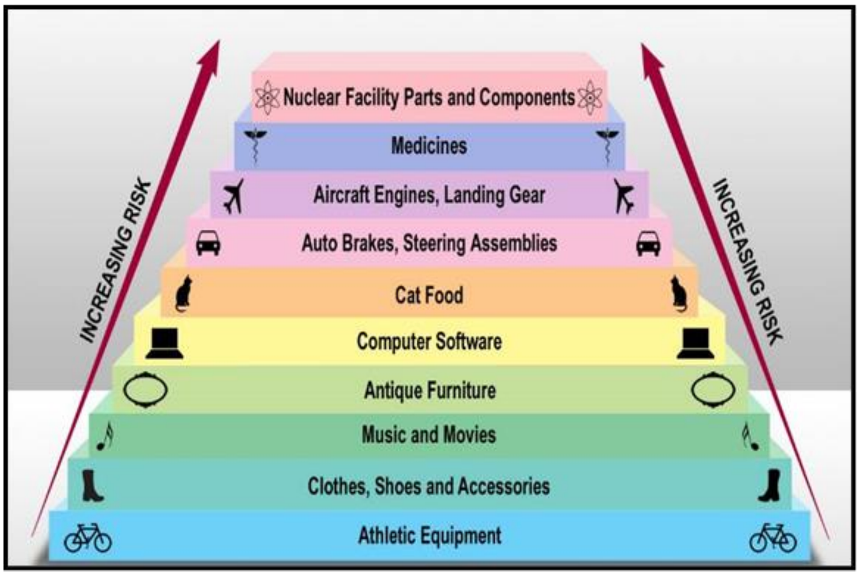


Figure 3. Counterfeiting Risk Profile

- Negatively impacts safety
- Damages the economy
- Victimises legitimate manufacturers and suppliers
- Causes loss of customer confidence
- Compounds the product liability issue
- Promotes unacceptable working practices such as human trafficking, child exploitation and organised crime

[2] https://www.oecd.org/en/about/news/press-releases

4.1 International Experience

Experience shows that CFSI include a wide range of items, such as threaded high strength fasteners, piping, other mechanical components, electrical and electronic components and individual's personal training, qualifications and certifications. Bulk materials and chemicals can also be of concern, including those provided to suppliers by their sub-suppliers. Recently, there have been documented cases within the nuclear industry related to fraudulent material, testing certifications, qualifications and fraudulent records associated with the delivery of services – see Appendix for further details.

CFSI are more likely to appear when:

- There is significant financial benefit for the counterfeiter
- The items are difficult to verify or not typically verified
- Procurement requirements (technical specifications) are poorly defined
- Weaknesses in processes and systems present opportunities for fraudulent activities to take place
- Methods or criteria for verifying that procurement requirements are met are inadequate
- Supply chains are complex
- Urgent replacement of an item is required (i.e. there are schedule pressures)
- The item is supplied from a single source with unreliable or unverified performance
- The item is difficult to source, or specifications/requirements are difficult to achieve
- Poor visibility of the entire supply chain
- There is not a strong nuclear safety culture within the organisations involved

4.2 CFSI Impact Statement on the UK

Recent analysis[3] reveals that the global trade in goods infringing UK Intellectual Property (IP) rights has adverse implications. Originating overseas and destined for the United Kingdom, these products have a total value of GBP £12.1 billion. In the United Kingdom, this trade led to 1.4% in forgone sales, around 20000 job losses, and 0.23% of public revenue forfeited. Analysis shows that 53% of imported counterfeit and pirated goods into

[3] Stryszowski, P. and M. Gaudiau (2024), "Trade in counterfeit products and the UK economy", *OECD Trade Policy Papers*, No. 287, OECD Publishing, Paris,

the UK in 2021 were sold to consumers who knew they were buying fake products, with the remaining share purchased unwittingly.

Overall, infringement of UK trademarks, patents and Intellectual Property in global trade amounted to more than 22000 job losses, which is equivalent to 1.1% of the total number of employees in the UK manufacturing sector.

4.3 CFSI Warning Signs

CFSI by their very nature are hard to identify due to the intention to deceive. SZC and the supply chain must recognise the warning signs of potential CFSI including but not limited to:

- Shorter lead-times in comparison to other suppliers
- Lower than expected or overly competitive prices
- Sudden availability of obsolete or difficult to source items, or production of missing certification or records (including Lifetime Quality Records – LTQR)
- Changes in items appearance, weight, colour, labelling etc.
- Differences in packaging
- Spelling mistakes or typographical errors on documentation, records or item labelling
- Use of signature images or typed signatures on documentation, Inspection and Test Plans (ITP), certification or records/LTQR, when wet-signed or controlled digital e-signatures are expected
- 'Grey Market' goods – purchasing from unauthorised vendors or distributors
- Items that are 'too good to be true'

5 SZC CFSI POLICY

The SZC Anti-CFSI Policy identifies three fundamental principles in the mitigation of CFSI risk to the SZC Project:

- **Prevention**
- **Detection**
- **Resolution**

These are supported by the SZC Anti-CFSI arrangements which identify seven lines of defence that encompass Prevention, Detection and Resolution:

- Supplier Selection – Pre- Qualification, Tendering and Contract Award
- Contract Terms and Conditions – including Contract Baseline Library specifications
- Supplier Oversight – Risk-based auditing and surveillance to ensure effective flow down/cascade of requirements across all tiers
- Industrial Affiliations – membership and collaboration
- Training and Awareness – clarity and understanding of requirements
- Investigation – Incident investigation, Root Cause Analysis (RCA) and resolution processes
- Communication – Event Alerts and Quality briefings

The objectives of these arrangements can only be achieved by the commitment of all members of the SZC Project together with supply chain partners, contractors and suppliers seeking the highest standards of quality to ensure we get it Right First Time (RFT).

5.1 Collaboration

The SZC Project is committed to working together with its supply chain to deliver excellence in everything it does; this is achieved by having the right people, with the right skills in place to deliver products and services at the right time.

This requires the SZC Project and its suppliers to develop a culture of supportive relationships, working together to deliver to the required standards, identifying opportunities to improve performance and resolving issues promptly in line with the SZC Quality Commitments (**Fig. 4**) through:

- Open and honest communication
- Raising queries and concerns as they become known - at all levels and tiers, throughout the supply chain

- Checking for understanding and compliance at all levels and tiers of the supply chain – ensuring that specifications, requirements, and identified processes are flowed down, understood, and met
- Conduct training on a regular basis to increase awareness of CFSI risk and mitigation
- Identifying and building on best practice
- Utilising Operational Experience (OPEX)/Retour d'Experience (REX) to share learning, prevent issues and improve performance
- Documented process for lessons learnt to demonstrate improvement and repeat success
- Implementing robust corrective and preventive actions

Figure 4. SZC Quality Commitments

To ensure the right culture is established and collaborative behaviours are in place to support the fight against CFSI, suppliers should ensure that all necessary personnel are aware of CFSI as a topic and the risks it presents to their business and the wider nuclear industry. This may include dedicated CFSI training, awareness briefings or use of industry standard training materials, videos or courses.

5.2 SZC Support

To aid the SZC supply chain partners in the management of CFSI, SZC offers CFSI support and collaboration to ensure suppliers are aware of the CFSI topic and SZC CFSI requirements are fully understood. SZC can also direct suppliers to industry-wide information on CFSI. Suppliers seeking help and support on CFSI matters should liaise directly with their SZC Quality contact.

To supplement SZC Quality support, the following materials are referenced as additional information available to the supply chain, some of which are included in the Appendix:

(1) ISO 19443:2018 - Quality management systems — Specific requirements for the application of ISO 9001:2015 by organizations in the supply chain of the nuclear energy sector supplying products and services important to nuclear safety (ITNS)

(2) IAEA Managing Suspect and Counterfeit Items in the Nuclear Industry [IAEA-TECDOC-1169]

(3) IAEA Managing Counterfeit and Fraudulent Items in the Nuclear Industry [NP-T-3.26]

(4) Nuclear Energy Agency Regulatory Oversight of Non-conforming, Committee on Nuclear Counterfeit, Fraudulent and Suspect Items Regulatory Activities [NEA/CNRA/R(2012)7]

(5) Nuclear Energy Agency Operating Experience Report: Counterfeit, Committee on Nuclear Fraudulent and Suspect Items Regulatory Activities [NEA/CNRA/R(2011)9]

(6) World Nuclear Association - Supply Chain Working Group: Countering Counterfeit, Fraudulent and Suspect Items in the Nuclear Supply Chain

(7) Nuclear Institute: Safety Directors Forum – Supply Chain Quality Working Group: COUNTERFEIT, FRAUDULENT AND SUSPECT ITEMS: The Risks of CFSI and how to identify dangerous goods and services.

(8) Kings College London - Securing the Nuclear Supply Chain: A Handbook of Case Studies on Counterfeit, Fraudulent and Suspect Items

(9) Office of Nuclear Regulation (ONR) TAG 77 – Supply Chain Management Arrangements for the Procurement of Nuclear Safety Related Items or Services [NS-TAST-GD-077]

6 SZC CONTRACT TERMS AND CONDITIONS

Suppliers are contractually required to have robust processes in place to initially assess a prospective supplier and then to monitor their work to ensure that it continues to meet the quality, schedule, safety and price requirements that both they and SZC stipulate.

Suppliers are also required to flow those same standards down to their supply chain tiers and to confirm their understanding.

- All SZC contracts and purchase orders contain requirements that goods and materials will be new not reconditioned, used, or repaired unless otherwise specified
- Contractors are obliged to ensure that all goods and materials are supplied with original documentation and accompanied with transposed certificates of origin
- Contracts contain conditions that prohibit delivery of CFSI and ensure that suppliers are aware of their accountability for providing the correct items and the consequences for supplying CFSI

Contracts include provisions for retaining part, or all of the fee until goods and materials are received that conform to SZC technical requirements, and hold contractors accountable for replacing CFSI with genuine items at their own expense.

Contracts are supported by Contract Baseline Library (CBL) specifications which include requirements in support of CFSI management such as Sub-Contractor Approval and Management [CBL100100770] to ensure capable and competent sub-contractors are selected and the sub-supply chain is mapped and known, SZC Culture Requirements [CBL100100853] to ensure the right behaviours are in place to prevent detect, report and resolve CFSI cases, and the SZC General Quality Assurance Specification (GQAS) [CBL100100684].

6.1 SZC General Quality Assurance Specification (GQAS)

The SZC General Quality Assurance Specification (GQAS) [CBL100100684] includes specific clauses and requirements on CFSI. The GQAS is aligned to the requirements of ISO 19443:2018 and requires suppliers to have robust arrangements in place to manage and mitigate the risk of CFSI in their scope of supply, including any sub-contracting to their own supply chains.

6.2 SZC – NEC Contract Clause

“The Contractor warrants that no Plant and Materials or Equipment that are counterfeit in whole or in their component parts or constituents are or will be used or incorporated into the works. Without prejudicing the Employer's other rights and remedies, the Contractor promptly replaces any suspect or counterfeit Plant and Materials or Equipment with such genuine Plant and Materials or Equipment that are acceptable to the Supervisor.

Any suspect or counterfeit Plant and Materials or Equipment may be impounded by the Supervisor and provided to the relevant authorities for investigation. The Contractor is liable for all costs relating to the removal, replacement and/or impoundment of such items and the Employer is entitled to withhold payment for such items pending the results of any investigation by the Employer or any relevant authority. The Contractor has no entitlement to a compensation event as a result of such removal, replacement and/or impoundment except in accordance with this clause"

6.3 SZC – FIDIC Contract Clause

"The Contractor warrants that no materials or Goods that are counterfeit in whole or in their component parts or constituents shall be Supplied. Without prejudicing the Employer's other rights and remedies, the Contractor shall promptly replace any suspect or counterfeit materials or Goods with such genuine materials or Goods that are acceptable to the Employer.

Any suspect or counterfeit materials or Goods may be impounded by the Employer and provided to the relevant authorities for investigation. The Contractor shall be liable for all costs relating to the removal, replacement and/or impoundment of such items and the Employer shall be entitled to withhold payment for such items pending the results of any investigation by the Employer or any relevant authority. The Contractor shall not be entitled to any extension of time for any delay or payment of any Cost arising in relation to the Supply as result of the removal, replacement and/or impounding of suspect or counterfeit materials or goods or their investigation, unless any investigation undertaken by the Employer or relevant authorities demonstrate that the materials or Goods in question were not counterfeit".

7 REPORTING OF CFSI

The supplier shall determine the rigour of inspection and test requirements for the acceptance of items. This shall be commensurate with the risk of the items being counterfeit or fraudulent and the criticality of the item in relation to safety and performance. Suppliers should consider sampling regimes, review of documentation such as material certification, and the use of Positive Material Identification (PMI) appropriate to the nature of items received.

If an item is suspected of being counterfeit or fraudulent at any point in production or service provision, then appropriate control via segregation or quarantining should be in place while additional testing is considered to confirm if the item is counterfeit or fraudulent.

Control of CFSI

The supplier shall establish arrangements to:

- Control suspected counterfeit or fraudulent items to prevent unintended use or re-entry into the supply chain – this may include physical identification and/or quarantining in a bonded secure location
- Ensure suspected counterfeit or fraudulent items are not returned to the sub-supplier unless under controlled circumstances for validation or testing
- Ensure that items confirmed as being counterfeit or fraudulent do not re-enter the supply chain and are not returned to the sub-supplier – this may include destruction of the item

Reporting of CFSI

The supplier shall establish arrangements to ensure that occurrences of counterfeit and fraudulent items are openly reported to:

- The customer and SZC
- The supplier of the item
- The owner of the Intellectual Property Rights of the genuine material
- Appropriate information/data gathering organisations including, but not limited to:
  - Anti-Counterfeiting/Fraud Forum
  - National Law Enforcement Authorities[4]

[4] Note: In the UK, occurrences of counterfeit material are also reported to the Police & Trading Standards

What to do if you discover Suspect Items

- Never assume that you will always get the correct product
- Always request and check and/or validate the original certification
- For raw materials, verify its chemical and mechanical properties – e.g. via independent testing or use of a Positive Material Identification (PMI)
- What should I do if I suspect CFSI – Isolate/quarantine the item, to avoid its inadvertent or unintended use and contact your company's Quality department to allow them to investigate further and raise a Non-Conformance Report (NCR)

8 SUMMARY AND CONCLUSION

CFSI are an ever-increasing problem for industry in general. Nuclear facilities need to be made aware of and have processes in place to prevent, detect and report CFSI. These processes should include ensuring good knowledge among supply chain participants, putting processes in place to ensure effective cascade of requirements down the supply chain, and monitoring and evaluating supply chain performance.

Generally, counterfeiters and fraudsters go after recognised, high demand items to maximise their profit. CFSI of concern to nuclear facilities include items that look nearly identical to original items but contain substandard, poorly assembled or aged components or material. Such items can be difficult to detect by standard industrial quality control inspections but can cause catastrophic failures or loss of functional capability. The infiltration of CFSI into industry could also lead to loss of legitimate firms from the marketplace, with an associated loss of jobs and revenue. Nuclear supply chains might have a decreased ability to deliver genuine products when needed, with potential negative impacts on facility reliability, economics and safety.

We must all be vigilant to the risk of using CFSI and question anything that appears doubtful, whether a SZC employee or a SZC supplier in any tier of the supply chain. We must ensure that the process of initially selecting and approving a supplier is robust and apply monitoring of the supply chain to ensure continuing understanding and compliance with contract and specification requirements.

CFSI are an ongoing and growing problem, not just within manufacturing but also in our daily lives.

SZC employees, contractors and the supply chain all have a role to play in ensuring that CFSI do not infiltrate the SZC supply chain, nor make it onto the SZC construction site.

9 TERMS AND DEFINITIONS

Term / Abbreviation	Definition
CBL	Contract Baseline Library (specifications)
CFSI	Counterfeit, Fraudulent and Suspect Items
Counterfeit Items	items intentionally manufactured, refurbished or altered to imitate an original or legitimate product without authorisation to pass themselves off as genuine.
CQAP	Contract Quality Assurance Plan
FIDIC	Fédération Internationale des Ingénieurs-Conseils
Fraudulent Items	items are intentionally misrepresented with intent to deceive, including items with incorrect identification or false/inaccurate certifications
GQAS	General Quality Assurance Specification
HPC	Hinkley Point C
Items	Although ‘I’ stands for item, it can be a product but can also be a service.
IAEA	International Atomic Energy Agency
IP	Intellectual Property
ITP	Inspection and Test Plan (or FUD)
LFE	Learning From Experience (see OPEX and REX)
LTQR	Lifetime Quality Record
Non-conforming item	Items that do not conform to the agreed specification or have been rejected by the purchaser supply chain or others
NCR	Non-Conformance Report
NDT	Non-Destructive Testing
NEC	New Engineering Contract – developed by the Institute of Civil Engineers
OEM	Original Equipment Manufacturer
ONR	Office for Nuclear Regulation
OPEX	Operational Experience (also see LFE & REX)
PCN	Personnel Certification in Non-Destructive Testing
PMI	Positive Material Identification
RCA	Root Cause Analysis
REX	Retour d'Experience (also see LFE & OPEX)
RFT	Right First Time
SDF SCQWG	Safety Directors Forum (Nuclear Institute) Supply Chain Quality Working Group
Suspect Items	items about which there is an indication by visual inspection, testing, or other preliminary information that it may not conform to accepted standards, specifications and/or technical requirements and there is a suspicion that the item may be counterfeit, or fraudulent
SQEP	Suitably Qualified and Experienced Personnel
SZC	Sizewell C

APPENDIX A - CFSI EXAMPLES

A.1 Example of CFSI in certification - Impact on UK-EPR

Problem statement

Hinkley Point C (HPC) received a notification that an NDT (Non-Destructive Testing) company (NDT International PTE Ltd) had its certification withdrawn due to falsified certificates being generated.

Supporting information

The British Institute of Non-Destructive Testing (BINDT), who manage Personnel Certification in Non-Destructive Testing (PCN) qualifications for NDT engineers had withdrawn the approvals and certification relating to NDT International PTE Ltd.'s approval to examine, qualify and certify NDT engineers.

All NDT engineers who had been certified by NDT International PTE Ltd have had their PCN certification withdrawn and were made to re-test. (It should be noted that not all certificates issued were falsified).

Acting on suspicions raised about the SQEP of certain NDT engineers BINDT carried out 'secret shopper' investigations and found that PCN qualifications were being falsified by NDT International PTE Ltd.

BINDT stated that this has affected around 3200 engineers (roughly 6500 certificates).

There was obviously a concern about any material that was tested by engineers certified by NDT International PTE Ltd.

Further information can be found on the BINDT web site[5].

Impact statement

Possible impact on the supply chain of the Nuclear industry in general.

Impact mitigation

Review and validation of supplied certificates at receiving inspection.

[5] http://www.bindt.org/Certification/pcn-news/

A.2 Example of CFSI in certification - Impact on EDF Nuclear Operations Ltd

Problem statement

In June 2015, EDF Nuclear Operations became aware of OPEX from Ontario Power Group of a UK supplier supplying valve products with misrepresented test certification.

Subsequent investigations by EDF Supply Chain confirmed that a material supplier to this UK valve supplier falsified test certification relating to American Society of Mechanical Engineers (ASME) material utilised in the manufacture of valves.

Supporting information

The issue was first known to the supplier in 2013 (from a Korean customer), although the matter was not made public until March 2015. Customers were notified progressively from April 2015.

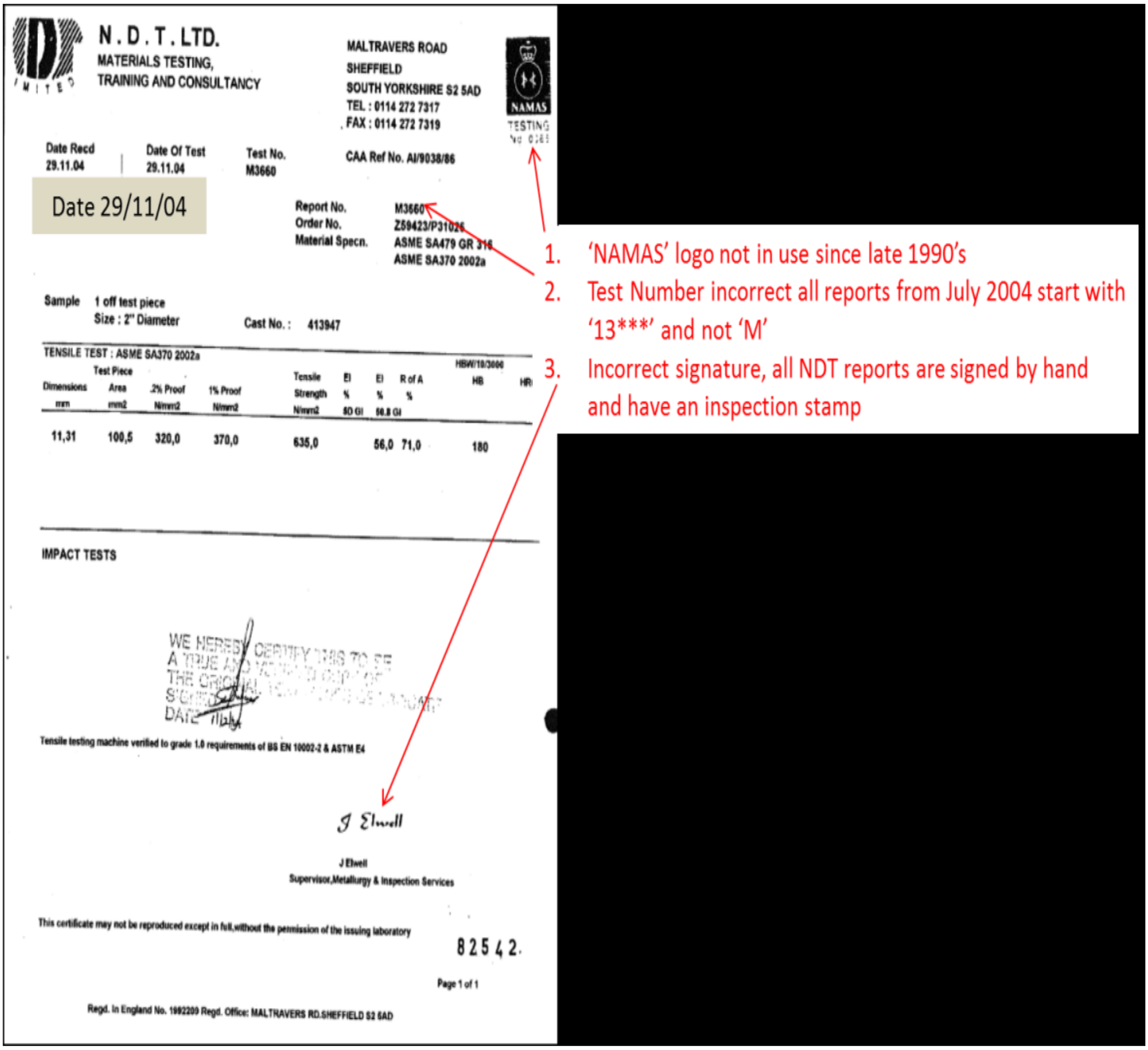
Following the OPEX/REX, EDF contacted the supplier and received a draft letter with preliminary information containing details of the falsification issue along with the number and type of affected valve components supplied to EDF. The formal letter was received from them in July 2015.

Impact statement

Over 1,000 valve components had been supplied to EDF with potentially misrepresented certifications, although after a filtering exercise, the supplier concluded that only 87 valves required more detailed justifications.

The affected valves and components include pressure parts (bonnets, body connectors, etc.) and valve internals (discs, pistons, valve stems, etc.).

The falsification of records took place for some years between 2001 and 2011.



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TESTING

Date Recd	Date Of Test	Test No.	CAA Ref No. AI/9038/86
29.11.04	29.11.04	M3660	

Date 29/11/04

Report No. M3660
Order No. Z59423/P31026
Material Specn. ASME SA479 GR 316
ASME SA370 2002a

Sample 1 off test piece
Size : 2" Diameter
Cast No. : 413947

TENSILE TEST : ASME SA370 2002a

Test Piece Dimensions mm	Area mm2	.2% Proof N/mm2	1% Proof N/mm2	Tensile Strength N/mm2	El % 50 GI	El % 50.8 GI	R of A %	HBW/10/3000 HB	HRB
11,31	100,5	320,0	370,0	635,0		56,0	71,0	180	

IMPACT TESTS

WE HEREBY CERTIFY THIS TO BE A TRUE AND ... COPY OF THE ORIGINAL ...
DATE

Tensile testing machine verified to grade 1.0 requirements of BS EN 10002-2 & ASTM E4

J Elwell

J Elwell
Supervisor,Metallurgy & Inspection Services

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Page 1 of 1

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NDT Certificate - Impact on EDF Nuclear Operations Ltd

A.3 Example of CFSI in certification - Impact on Sizewell C

Problem statement

Example of modified chemical analysis results, residual stress and hardness measurements – potentially impacting upon steam generator parts and reactor pressure vessels

During May 2022, a whistle-blower made allegations of inappropriate activities within the Japan Steel Works (JSW) organisation stretching back to 1998. JSW published reports of irregularities at its Muroran plant in Japan. Specifically, the modification of chemical analysis results, residual stress measurements and hardness measurements[6].

Supporting information

JSW manufactures numerous components for the nuclear industry world-wide. In particular reactor pressure vessel (RPV) and primary circuit forgings, including for the HPC, SZC and EPR2 reactors. Manufacturing stopped and certification was withdrawn. A special investigation committee, including lawyers external to JSW, was set up and tasked with conducting an in-depth investigation.

EDF SA France informed French nuclear regulator Autorité De Süreté Nucléaire (ASN) that irregularities were reported. Initially, the nuclear sector was not directly concerned. However, to guarantee the situation for products intended for the nuclear sector, EDF SA set up a multi-company task force.

During November 2022, the special investigation committee report highlighted irregularities affecting twenty components of equipment intended for the nuclear industry, the oldest dating back to 2013 including 6 cases concerning EDF orders.

Impact statement

Manufacturing for SZC was put on hold and only restarted under strict controls after 6 months and full production and re-certification was only achieved some 18 months later.

[6] https://www.french-nuclear-safety.fr/information/news-archives/irregularities-in-equipment-manufacturing-at-japan-steel-works

A.4 Example of CFSI in certification - Impact on EDF SA and Areva Creusot Forge NP

Problem statement

In 2016, Areva NP highlighted irregularities in certain manufacturing files for nuclear pressure equipment manufactured by its Creusot Forge plant (Saône-et-Loire département). EDF SA and Areva NP conducted a manufacturing quality review in its Creusot Forge plant (Saône-et-Loire département), which revealed the existence of irregularities in certain manufacturing files[7].

Supporting information

The first investigations carried out on these files in 2016 identified 89 irregularities concerning EDF SA reactors in operation. Autorité De Süreté Nucléaire (ASN) who are the French Nuclear Regulator asked EDF SA to extend the review to all the manufacturing files for components forged in this plant.

The purpose of this extended review was to detect any deviation from either the technical baseline requirements chosen by the manufacturer, or the plant's internal requirements, or from the contractual or regulatory requirements applicable at the time of manufacture.

Starting in July 2017, EDF SA transmitted the results of the manufacturing file reviews for twelve reactors for which restart, following a refuelling outage was scheduled between

[7] https://www.french-nuclear-safety.fr/information/news-archives/irregularities-detected-in-areva-s-creusot-forge-plant

September and November 2017. At that stage, the review led to the detection of 601 conformity deviations. ASN analysed the results transmitted by EDF SA before it authorised the restart of these reactors.

Impact statement

This led to significant recovery actions from Areva NP, this involved the suspension of manufacturing and the loss of customer confidence.

Since this multinational inspection, the ONR has developed its intervention plans to ensure that the licensee has in place and implements adequate management and assurance arrangements to clearly demonstrate that all components are manufactured to the required standards.

SZC also has its own inspection and quality assurance programme to provide the required confidence that the components manufactured by Areva NP (now Framatome as of July 2018) for the SZC Project to meet those exacting standards.

A.5 Example of CFSI in certification - Counterfeit pipe failure at Datong power station

Problem statement

There was a steam line failure from a 300MW unit at the Datong Power Station Unit 2 in China. The investigation indicates that the pipe was actually manufactured by a Chinese pipe mill, then sold to another Chinese "pipe producer" who did not perform the specified heat treatment, then sold to a third “pipe company" that polished the OD and sized the ends. The pipe was then shipped to a company in Houston USA which certified it as “Made in the USA” and shipped it to the fabrication company, which then supplied it for installation on the Datong station.

Supporting information

The pipe was installed early to mid-2006 and had been in operation for less than six weeks when the rupture occurred. Tragically, two people died, and many were injured. These details have been confirmed by the Bechtel QA Manager of Power. Bechtel China has also conducted an investigation as the supply chain is complex due to the extent of how many agents, brokers, and mills are involved.

Impact statement

There is evidence to support that over 30 plants contain similar or other "fake pipe" all over China. Documented instances exist where this type of pipe has been transferred into the US market.

The Chinese government has called for a formal investigation and banned Chinese-made pipe for use in major power plant critical applications.

Counterfeit pipe failure at Datong power station

A.6 Example of CFSI - Counterfeit Yokogawa Transmitters

Problem statement

Yokogawa had become aware of several instances in which counterfeit field instruments, bearing the Yokogawa logo, had reached some of their customers. At first glance, they were nearly indistinguishable with a semblance of functionality and interface that mimics the correct product.

Supporting information

A thorough investigation confirmed that these counterfeit instruments were being produced by unauthorised manufacturers who had gone to great lengths to imitate Yokogawa products. Performance test results showed that they are severely inferior in quality, and performance and that they pose a serious safety risk.

It is important to note that these counterfeit instruments carry tagging that indicates they are conformant, but they are not conformant for use in ExD (Explosive Devices) and other hazardous areas. They also do not meet the engineering and manufacturing requirements of regulatory bodies.

Impact statement

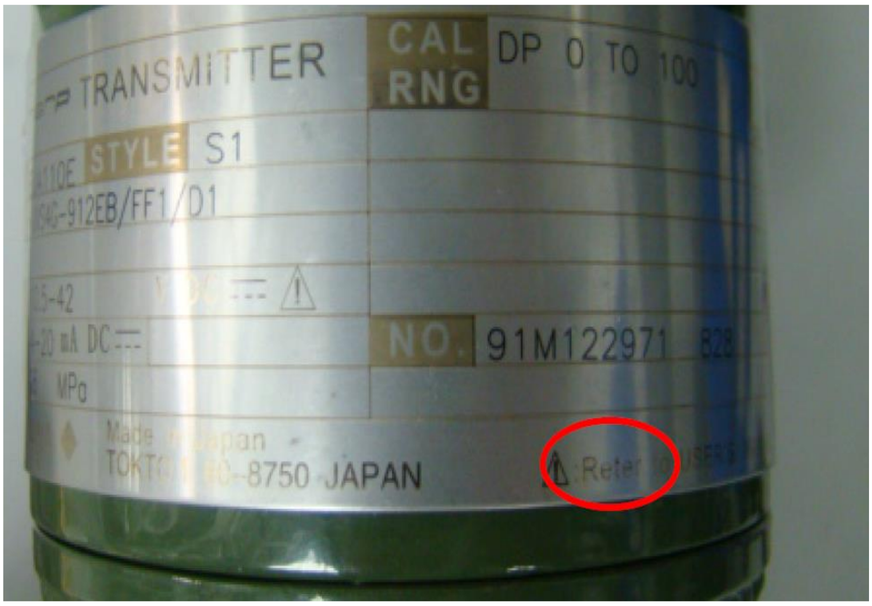
Instruments cannot be relied upon to operate as specified in all conditions.

Yokogawa are working with authorities to reduce or eliminate the delivery of counterfeit instrumentation to its customers.

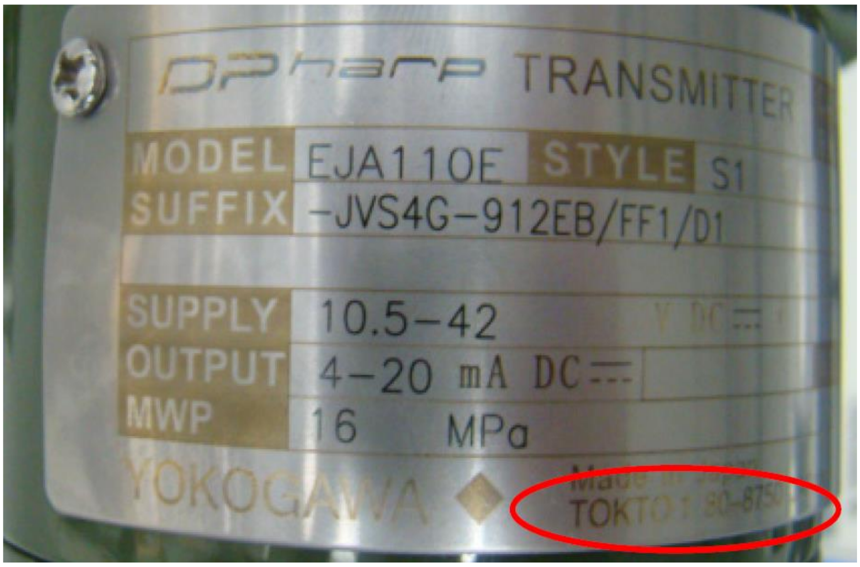
Counterfeit example of a Yokogawa Transmitter

Top Works Name Plate of a counterfeit has several typos:

1. “Reter” instead of “Refer”.



2. Some units will state “Tokto” instead of “Tokyo”.



3. Thread type identification letter stamp “N, A or M” is missing on counterfeits.


Yokogawa


Counterfeit

A.7 Example of CFSI in certification – Impact on Kobe Steel

Problem statement

Although the problem of companies using substandard materials usually involves small/medium enterprise (SMEs), the most recent incident regarding counterfeiting material involves a large corporation Kobelco (Kobe Steel Ltd) based in Japan.

Supporting information

Employees at Kobe Steel Ltd had falsified Material Test Certificates (MTC's) to indicate the materials met the customer specifications when in fact they did not. The fraud lasted over a decade and had cost the company millions. Two days after the announcement, Kobe Steel shares price fell 30% in value, wiping $700 million USD off its market value.

The scandal has forced some of Japan's best-known manufacturers to confirm the safety of products sourced from Kobe Steel.

Toyota, Nissan and US carmaker General Motors are among about 500 affected companies, and Hitachi said it had used Kobe Steel parts in trains built for the UK market.

Impact statement

The chief executive of Kobe Steel has said a deepening scandal over false inspections data may have spread beyond Japan and conceded that his company now had "zero credibility".

Kobe Steel Ltd admitted its data fraud has been going on nearly five decades and revealed new cases of cheating, highlighting the challenges facing the 112-year-old company mired in compliance failures and malfeasance, throwing global supply chains into turmoil.

A.8 Examples of CFSI within the steel industry

Top ten list of fake steel products

A Construction Industry Institute (CII) study[8] had found that raw, substandard steel and associated products were the most counterfeited commodity among the materials (within the construction industry). CII outlined a top ten list of the most counterfeited commodities:

- Steel
- Fasteners
- Valves
- Pipe
- Circuit breakers
- Rotating equipment parts
- Electric equipment
- Pipe fittings
- Pressure vessels
- Cement

[8] Waging War on Suspect/Counterfeit Parts | THE RAM REVIEW